Revision | **Source** Lecture review slide | **File** 10 / 11

RV

The review self-study exercises, fully solved

تمارين المراجعة محلولة بالكامل

These are the three exercises Dr. Al-Hadhrami closes the chapter with, set for self-study. They are solved here step by step.

الشرح بالعربي

هذه تمارين المراجعة الثلاثة في آخر ملف الدكتور. الفكرة المشتركة بينها: **ارسم مخطط فن واملأ المناطق الأربع بالأرقام** (A فقط، B فقط، المشترك، خارج الاثنين)، ثم يصبح كل فرع مجرد جمع لمناطق تراها أمامك. وتذكّر أن **neither … nor** تعني دائمًا $1 - P(A \cup B)$.

EX 1 · BURGERS AND PIZZA

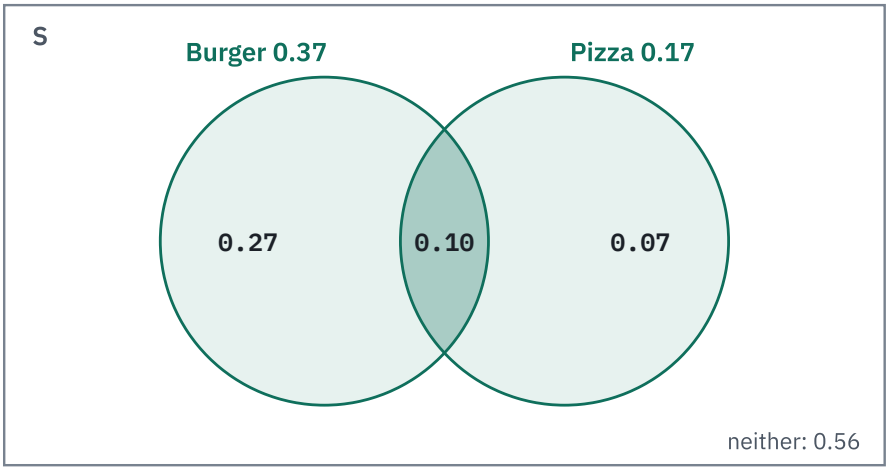
A study shows that about 37 percent of school children eat burger, 17 percent eat pizza, and 10 percent eat both burger and pizza. What is the probability that a randomly selected school child eats **neither** burger nor pizza?

1. **Extract:** P(B) = 0.37, P(Z) = 0.17, P(B ∩ Z) = 0.10.
2. **"Neither…nor" is the complement of the union.** First get the union — they overlap, so Rule 2:

   `P(B ∪ Z) = 0.37 + 0.17 − 0.10 = 0.44`

3. `P(neither) = 1 − P(B ∪ Z) = 1 − 0.44 =` **`0.56`**

Cross-check with a Venn diagram: burger only = 0.37 − 0.10 = 0.27; pizza only = 0.17 − 0.10 = 0.07; both = 0.10; outside = 1 − (0.27 + 0.07 + 0.10) = 0.56. ✔



Fill the overlap first (0.10), then subtract it from each total to get the “only” regions. The four regions add to 1, and “neither” is the region outside both circles.

املأ المنطقة المشتركة أولًا (0.10)، ثم اطرحها من كل مجموع لتحصل على «فقط». مجموع المناطق الأربع = 1، و«لا هذا ولا ذاك» هي المنطقة خارج الدائرتين.

EX 2 · SMART PHONE COLORS

If the probabilities are, respectively, 0.23, 0.15, 0.21, and 0.09 that a person purchasing a new smart phone will choose the color gold, silver, black, or white, what is the probability that a given buyer will purchase a new phone that comes in one of those colors?

1. A phone has exactly one color, so the four events are **mutually exclusive** → extended Addition Rule 1.
2. $P(\text{gold} \cup \text{silver} \cup \text{black} \cup \text{white}) = 0.23 + 0.15 + 0.21 + 0.09 =$ **0.68**

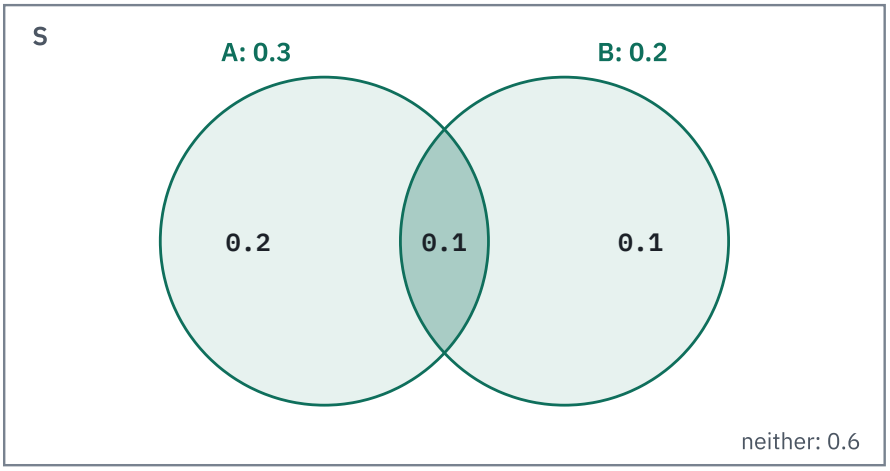
Read what this tells you: the total is 0.68, not 1 — so 32% of buyers choose some other color entirely. A question that follows up with "find the probability the buyer chooses a different color" is answered by 1 − 0.68 = 0.32.

EX 3 · SET ALGEBRA WITH PROBABILITIES

If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine: (a) $P(A^c)$ (b) $P(A \cup B)$ (c) $P(A^c \cap B)$ (d) $P(A \cap B^c)$ (e) $P((A \cup B)^c)$ (f) $P(A^c \cup B)$

Build the four-region picture first. A Venn diagram with two overlapping circles has exactly four regions, and every part of this question reads straight off it:

REGION	MEANING	PROBABILITY
A only	$A \cap B^c$	$0.3 - 0.1 = 0.2$
Both	$A \cap B$	0.1
B only	$A^c \cap B$	$0.2 - 0.1 = 0.1$
Neither	$(A \cup B)^c$	$1 - 0.4 = 0.6$
Total	S	**1.0 ✔**



The four regions of Ex 3. Every part of the question is a sum of regions: $A \cap B^c$ is the left region (0.2), $A^c \cap B$ is the right region (0.1), and $A^c \cup B$ is everything except the left region (0.8).

مناطق التمرين 3. كل فرع هو مجموع مناطق: A فقط = 0.2، B فقط = 0.1، و $A^c \cup B$ = كل شيء ما عدا منطقة A فقط = 0.8.

1. **(a)** $P(A^c) = 1 - P(A) = 1 - 0.3 =$ **0.7**
2. **(b)** $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.2 - 0.1 =$ **0.4**
3. **(c)** $P(A^c \cap B)$ = "B but not A" $= P(B) - P(A \cap B) = 0.2 - 0.1 =$ **0.1**
4. **(d)** $P(A \cap B^c)$ = "A but not B" $= P(A) - P(A \cap B) = 0.3 - 0.1 =$ **0.2**
5. **(e)** $P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.4 =$ **0.6**

6. **(f)** $P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B) = 0.7 + 0.2 - 0.1 =$ **0.8**
   Or from the diagram: everything except "A only" = 1 − 0.2 = 0.8 ✔

> **THE TECHNIQUE THAT MAKES THIS QUESTION TRIVIAL**
>
> Never attack set-algebra questions with formulas alone. Draw the two circles, fill in the four regions as numbers, and every sub-part becomes an addition of regions you can see. Parts (c), (d) and (f) take longer to write than to solve once the diagram exists.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved ← this file

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.